MoCafe v2.00 — A Monte-Carlo Dust Radiative-Transfer Code with Scattering and Thermal Emission: User Guide

Abstract

MoCafe is a parallel (Fortran 90 + MPI) Monte-Carlo radiative-transfer code for dust scattering and thermal emission. It transports photon packets over a wavelength grid through a dusty medium, scatters them off dust grains (Henyey–Greenstein phase function or a tabulated Mueller matrix), tallies the radiation field in every cell, and re-emits the absorbed energy as thermal dust emission through either the Lucy (1999) method coupled to the SEDUST grain library (equilibrium and stochastically heated grains, including PAH features) or the Bjorkman & Wood (2001) immediate-reemission method. It supports multiple stellar populations and galaxy models, an all-sky interior observer (for the Milky-Way case), and a Modified Random Walk for very optically thick regions. This guide covers installation, execution, the complete input reference, the dust-emission workflow, the output products, worked examples, and the underlying algorithms.

Contents

1. Overview

MoCafe follows photon packets emitted by one or more sources through a dusty medium. Packets scatter off dust grains (Henyey–Greenstein phase function or a tabulated Mueller matrix) and are absorbed and re-emitted as thermal radiation. At every interaction a contribution is "peeled off" toward each observer to build an image or spectral-energy distribution (SED).

Capabilities.

- **Multi-wavelength (SED) transport** (`par%use_sed`): one run covers the full UV-to-far-IR range on a log-spaced wavelength grid.
- **Mean intensity in each cell** $J_\lambda(x,y,z)$ (`par%save_jlam`) via the Lucy (1999) pathlength estimator.
- **Dust thermal emission** (`par%use_dustemis`) in two modes: *Mode 1* (Lucy + SEDUST, grain models astrodust / DL07 / Zubko) and *Mode 2* (Bjorkman & Wood 2001).
- **Galaxy models**: multiple stellar populations (`par%nsource`) with radially-exponential and sech^2 disks, log-spiral arms, and Sersic (alias-sampled) / boxy / bar bulges.
- **All-sky interior observer** (`par%allsky`) for the Milky-Way case.
- **Modified Random Walk** (`par%use_mrw`) plus Russian roulette for very optically thick media.
- **Python analysis tools** (`python/`): a format-agnostic reader and plotters for SEDs, temperature/power maps, J_λ, and all-sky maps.

The emission features are opt-in: a plain monochromatic scattering run uses none of them. The design and validation record is in `MoCafe_v2.00_PLAN.md`.

Units. In scattering-only mode MoCafe is unit-agnostic and the image is in `(luminosity unit)` $\mathrm{cm^{-2}\,sr^{-1}}$. In SED / emission mode a physical length scale is required (through `par%distance_unit`: one grid length unit equals one distance unit) and the source luminosity (`par%luminosity`) is in $\mathrm{erg\,s^{-1}}$, so intensities and dust temperatures are absolute.

2. Installation

2.1 Prerequisites

- A Fortran 2008 MPI compiler: Intel oneAPI (`mpiifort` or `mpiifx`) or GNU (`mpif90`). MoCafe is MPI-only.
- **CFITSIO** (required).
- **HDF5** 1.14+ with Fortran bindings built by the same compiler (optional; on by default).
- Python 3 with `numpy`, `astropy`, `h5py` (and `scipy` for the Illustris converter) for the analysis tools.

2.2 Building MoCafe

```
make                              # HDF5=1 (default) -> MoCafe.x
make HDF5=0                       # CFITSIO only
make HDF5_PREFIX=/usr/local/hdf5  # override HDF5 location
make F90=mpif90                   # force a compiler
make DEBUG=1                      # checked / traceback flags
```

The compiler is auto-selected `mpiifort` → `mpiifx` → `mpif90`. The build always passes `-cpp -DMPI` and (by default) `-DHDF5`. When you add a source file, append its object to `OBJS` in the `Makefile` in dependency order.

2.3 Building the SEDust library

The Lucy emission engine (Mode 1) links the SEDust library, kept self-contained under `SEDust/`. Build it once:

```
cd SEDust/sed && ./build_lib.sh    # -> SEDust/sed/lib/libsedust.a  (Intel)
```

The MoCafe `Makefile` links `SEDust/sed/lib/libsedust.a` automatically (variable `SEDUST_LIBDIR`). The SEDust module named `mathlib` is renamed to `sed_mathlib` to avoid a clash with MoCafe's own `mathlib`. The optics tables the default `astrodust` path reads ($\sim$26 MB) are committed to the repository, and `par%sed_workdir` defaults to blank so MoCafe auto-resolves the SEDust directory relative to the `MoCafe.x` executable at run time — a fresh checkout runs dust emission from any working directory with no path editing. Only the 17 MB D16 spheroid Q-library (used by a non-default graphite path) is kept out of git; run `SEDust/populate_data.sh` to fetch it, editing its source path (or setting `SEDUST_SRC`) to your own SEDust tree. Mode 2 (Bjorkman & Wood) needs only the extinction table and not SEDust.

3. Running MoCafe

```
mpirun -np <N> ./MoCafe.x <input>.in
```

The single argument is a Fortran namelist file. Every tunable quantity is a field of the global `par` structure and is set as par%<name> = <value> inside a `&parameters / ... /` block. Sample inputs and `run.sh` scripts live under `examples/`. The default of every parameter is the initializer of `params_type` in `src/define.f90`.

Reproducibility. A run is bit-identical when `par%iseed` is a fixed non-zero value, `par%use_master_slave` is `.false.` (equal-share), and the number of MPI ranks is fixed (`par%iseed` = 0 seeds from the clock/PID).

4. Input Parameter Reference

Parameters are grouped below. Types: `L` logical, `I` integer, `R` real, `S` string, `[ ]` array.

4.1 Simulation control

Parameter	Default	Description
`no_photons`	1e5	Number of photon packets
`no_print`	1e7	Progress-print interval (photons)
`iseed`	0	RNG seed (0 → seeded from clock/PID)
`use_master_slave`	.true.	MPI mode: master–slave vs. equal-share
`num_send_at_once`	10000	Batch size (master–slave)
`luminosity`	1.0	Total source luminosity [erg/s] (SED/emission mode)

4.2 Grid geometry (Cartesian)

Parameter	Default	Description
`nx, ny, nz`	1,1,11	Grid resolution
`xmax, ymax, zmax`	1.0	Half-sizes of the box (grid spans $[-\max, \max]$)
`rmax`	-999	If > 0 and not a cylinder, forces a cube of side $r_{\max}$ (sphere)
`nr`	-999	If > 1, sets nx=ny=nz=nr
`geometry`	' '	", 'sphere', 'cylinder', or 'Plummer' (auto-set from source)
`xyz_symmetry`	.false.	Exploit octant symmetry (disabled if any $n = 1$)
`z_symmetry`	.false.	Exploit reflection about $z = 0$
`xy_periodic`	.false.	Periodic in x, y

4.3 Grid type and media

Parameter	Default	Description
`grid_type`	'car'	'car' (Cartesian), 'clump' (clumpy), 'amr' (octree)
`amr_file`	' '	Generic AMR octree file (FITS/HDF5/text) for 'amr'
`dust_model`	'global_dgr'	AMR leaf opacity: 'global_dgr', 'from_file', 'laursen09'
`DGR, Z_global, Z_ref, f_ion_dust`	—	Dust-to-gas / metallicity parameters (AMR)
`clump_radius`	-1	Clump radius (clumpy medium)
`clump_f_cov / _f_vol / _N_clumps`	-1	Clump count specification
`clump_tau0 / _ndust / _nH`	-1	Opacity specification for each clump

The clumpy and AMR media are documented in `docs/MoCafe_clump.pdf` and `docs/MoCafe_amr.pdf`. The full dust-emission suite runs on both the Cartesian grid and the AMR octree (§5.8); the clumpy medium is scattering-only.

4.4 Distance unit

Parameter	Default	Description
`distance_unit`	'kpc'	'', 'pc', 'au', 'kpc'; sets `distance2cm`. In SED mode one grid unit = one distance unit (physical scale)

4.5 Source geometry (single source)

Parameter	Default	Description
`source_geometry`	'point'	'point', 'uniform', 'uniform_xy', 'gaussian', 'exponential', 'sech', 'exp_spiral', 'external_{sph,cyl,rec}'
`xs_point, ys_point, zs_point`	0.0	Point-source location
`source_zscale`	NaN	Vertical scale height (gaussian/exponential/sech/exp_spiral)
`source_rscale`	NaN	Radial scale length of the exponential disk
`radiation_angular_PDF_file`	' '	Angular PDF for external illumination
`ext_geometry`	' '	External-field boundary *when composing* an internal source with an external field: 'sph'/'cyl'/'rec' (blank auto-selects from geometry/rmax); the external-only run uses source_geometry='external_*' instead (§5.1)

Multiple sources are described in §5.5.

4.6 Source configurations: monochromatic vs SED

MoCafe runs one or more internal sources, an external isotropic radiation field, or an internal source composed with an external field, in either the monochromatic (scattering-only) mode or the SED mode (Table 6). Multiple internal sources (`par%nsource` > 1, §5.5) work in *both* modes, not only in SED; a single internal source uses the scalar `par%source_geometry`. The external-only field uses `source_geometry='external_{sph,cyl,rec}'`, while the composed internal + external case (§5.1) is switched on by adding an external field to a non-external internal source.

4.7 Dust and scattering (grey / monochromatic)

Parameter	Default	Description
`albedo`	0.3253	Single-scattering albedo
`hgg`	0.6761	Henyey–Greenstein asymmetry g
`cext_dust`	1.61e-21	Dust extinction per H nucleus [cm^2]
`scatt_mat_file`	' '	Mueller-matrix table (`data/mueller_*.dat`); enables Stokes

Parameter	Default	Description
`use_stokes`	.true.	Stokes polarization (auto-off without a matrix file)

In SED mode the wavelength-dependent albedo, g, and $C_{\rm ext}$ are read from `par%kext_file` and override these scalars (§5.1).

4.8 Optical depth and density

Parameter	Default	Description
`taumax`	-999	Radial optical depth, center → edge (sets the density scale)
`tauhomo`	-999	Optical depth of the volume-equivalent homogeneous medium
`density_file`	' '	3-D density cube (FITS/HDF5)
`reduce_factor`	1	Density-file down-sampling
`centering`	0	Density-file re-centering mode
`density_zscale`, `density_rscale`, `density_powerlaw_index`	NaN	Analytic density modulations

In SED mode `par%taumax`/`par%tauhomo` are defined at the reference wavelength `par%lambda_ref`.

4.9 Peel-off observers

Parameter	Default	Description
`nxim, nyim`	129	Image pixels
`dxim, dyim`	NaN	Pixel size [deg] (auto to cover the grid if unset)
`distance`	NaN	Observer distance; renormalizes (obsx,obsy,obsz)
`inclination_angle, position_angle, phase_angle`	NaN	Observer angles [deg] (arrays)
`alpha, beta, gamma`	NaN	Euler angles [deg] (alternative)
`obsx, obsy, obsz`	NaN	Explicit observer direction (alternative)

Up to `MAX_OBSERVERS = 181` observers per run. Angle conventions are in `docs/MoCafe_Geometry.pdf`.

4.10 Output control

Parameter	Default	Description
`file_format`	'hdf5'	'hdf5' or 'fits'; authoritative for the container

Parameter	Default	Description
`out_file`	' '	Output base name (from the input name if empty)
`out_bitpix`	-32	FITS bitpix (−32 float32, −64 float64)
`save_direc0`	.false.	Also write the unattenuated direct image
`sightline_tau`	.false.	Also write a sight-line optical-depth map for each pixel

5. Dust Thermal Emission

The emission workflow is: *SED transport* (panchromatic scattering) → J_λ *tally* → *dust emission* → observer SEDs. Enable it with `par%use_sed`, and add `par%save_jlam` and `par%use_dustemis` for the emission.

5.1 Multi-wavelength (SED) mode

Parameter	Default	Description
`use_sed`	.false.	Enable panchromatic (multi-wavelength) scattering transport
`nlambda`	128	Number of log-spaced wavelength bins
`lambda_min, lambda_max`	0.0912, 2000	Wavelength range [μm]
`lambda_ref`	0.55	Reference wavelength [μm] for the grid opacity and τ
`kext_file`	' '	Table λ, ϖ, $\langle\cos\rangle$, $C_{\rm ext}/{\rm H}$
`source_spectrum`	' '	Two-column source spectrum λ, L_λ (columns set by spectrum_type)
`spectrum_type`	'shape'	Column units of every source spectrum file (below)
`tstar`	-999	Planck source temperature [K] (if no spectrum file)
`save_jlam`	.false.	Tally J_λ in each cell; *opt-in*, required by the `'lucy'` dust emission (§5.)
`use_dustemis`	.false.	Add dust thermal emission (§5.); **leave both this and save_jlam off for scattering only** over the wavelength range

Wavelength-dependent extinction is applied as a grey rescaling for each photon

$$s_{\rm ext}(\lambda) = \frac{C_{\rm ext}(\lambda)}{C_{\rm ext}(\lambda_{\rm ref})}, \tag{1}$$

so the grid stores only the reference-wavelength opacity and every optical depth is multiplied by $s_{\rm ext}$ (the same device as the Jonsson 2006 τ scan). All three ray tracers (Cartesian, clumpy, AMR) are therefore shared with the monochromatic path, and a run with $s_{\rm ext} = 1$ is bit-identical to the mono code.

The source spectrum is either a Planck function at `par%tstar` or a two-column file λ, L_λ (`par%source_spectrum`, or `par%src_spectrum(`i`)` for each component). `par%spectrum_type` fixes the column units and the shape-vs-absolute treatment: `'shape'` (default, and always for a Planck source) uses only the shape as the wavelength-sampling distribution

Table 6: Supported source configurations: monochromatic scattering vs. SED, and single / multiple internal / external / composed sources.

Source configuration	Monochromatic (scattering only)	SED (spectrum $\rightarrow$ dust emission)
Single internal source (point / uniform / gaussian / disk / bulge)	supported	supported
Multiple internal sources (`par%nsource` > 1)	supported	supported
External isotropic field (`source_geometry='external_*'`)	supported	supported
Internal source(s) + external field composed in one run	supported	supported

while the absolute scale is set by `par%luminosity` / `par%src_lum` (legacy behavior). A physical type is *absolute*: `'per_um'` ($\lambda[\mu$m], L_λ [erg s^{-1} μm^{-1}]), `'per_ang'` (λ[Å], L_λ [erg s^{-1} Å^{-1}]), `'per_hz'` (ν[Hz], L_ν), or `'per_ev'` (E[eV], L_E); the reader converts to L_λ per μm with the Jacobians $L_\lambda = L_\nu\, c/\lambda^2$ and $L_\lambda = L_E\, hc/\lambda^2$. For an absolute type, when `par%luminosity` / `par%src_lum` is left unset the source luminosity is derived from the file integral over the wavelength grid; when set, the file is rescaled to it. The helper `python/make_source_spectrum.py` writes ready-to-use tables under `data/`: single stellar populations from FSPS (`spec_{young,old}_fsps.dat`) and Starburst99 (`spec_{young,old}_sb99.dat`), and the Mathis et al. (1983) Milky-Way interstellar radiation field (`isrf_mathis.dat`). The `examples/galaxy` models use the FSPS young/old pair and `examples/milkyway` uses the Mathis field.

External illumination. SED mode also supports the external geometries (`external_sph`, `external_cyl`, `external_rec`). The external field is an isotropic mean intensity, given either as a shape (`par%ext_spectrum` file or `par%ext_tstar` Planck) scaled by the band mean intensity `par%ext_intensity` [erg s^{-1} cm^{-2} sr^{-1}], or as an absolute `par%ext_spectrum` file whose J is derived from the file integral when `par%ext_intensity` is unset. In an `par%ext_spectrum` file column 2 is the mean-intensity density J_λ in the `par%spectrum_type` units. The spectrum priority is `par%ext_spectrum` file $>$ `par%ext_tstar` Planck $>$ the global `par%source_spectrum`/`par%tstar`. The luminosity entering the grid is

$$L = \pi J A_{\rm surface}, \tag{2}$$

with $A_{\rm surface}$ the illuminated boundary area: $4\pi\,$(`par%rmax` $\cdot$ `par%distance2cm`)^2 for `external_sph`, the six box faces for `external_rec`, and the side plus both end caps for `external_cyl`. This L sets `par%luminosity` for the absolute image normalization.

Composition (internal source + external field). An internal source (or sources) and an external isotropic field can illuminate the medium in one run. It is activated when a non-external internal source is present (`par%nsource` $\geq$ 1, `par%source_geometry` not an `external_*` value) together with an external field — `par%ext_intensity` > 0 in the monochromatic mode, or `par%ext_intensity` > 0 / `par%ext_spectrum` / `par%ext_tstar` in SED mode. `par%ext_geometry` (`'sph'`/`'cyl'`/`'rec'`; blank auto-selects from `par%geometry`/`par%rmax`) fixes the boundary the external field enters through when composing; the external-only run keeps `source_geometry='external_*'` instead. With the internal total $L_{\rm int}$ (a single `par%luminosity`, or $\sum_i$ `src_lum`(i) for several sources) and the external luminosity

$$L_{\rm ext} = \pi J A_{\rm surface}, \qquad L_{\rm tot} = L_{\rm int} + L_{\rm ext}, \tag{3}$$

where J = `par%ext_intensity` (or J from `par%ext_spectrum` in SED mode) and $A_{\rm surface}$ is the same illuminated-boundary area as above, each photon's origin is drawn internal-vs-external in proportion to $L_{\rm int} : L_{\rm ext}$ and every packet carries $L_{\rm tot}/N_{\rm phot}$. `par%luminosity` is set to $L_{\rm tot}$ for the output normalization, the run reports `TOT_LUM` $= L_{\rm int} + L_{\rm ext}$, and the composed image equals (internal-only) + (external-only) within Monte-Carlo noise. Composition is incompatible with Stokes polarization and with the $(a, g)/\tau$ scans; pre-existing external-only and internal-only runs are unchanged.

Restrictions: SED mode is incompatible with Stokes polarization and with the $(a, g)/\tau$ single-run scans.

5.2 Mean intensity J_λ in each cell

Parameter	Default	Description
`save_jlam`	.false.	Tally J_λ and write `<base>_jlam`

The mean intensity is accumulated with the Lucy (1999) pathlength estimator: each path segment of length $d\ell$ contributes $w L_{\rm pk}\, d\ell$ to its (wavelength-bin, cell) slot, and

$$J_\lambda(\text{cell}) \;=\; \frac{1}{4\pi V_{\rm cell}\, \Delta\lambda\, d_{\rm cm}^2} \sum_{\rm paths} w L_{\rm pk}\, d\ell, \tag{4}$$

where $L_{\rm pk}$ is the packet luminosity [erg/s] and $d_{\rm cm}$ the distance-unit conversion. The forced first-flight is tallied analytically (a zero-variance direct term). A built-in energy check reports the absorbed luminosity from the tally against an independent event counter. Memory is $\approx n_\lambda \times n_{\rm cell} \times 8$ bytes per MPI rank ($n_{\rm cell} = n_x n_y n_z$ on the Cartesian grid, or the number of leaves on the AMR octree; see §5.8).

5.3 Mode 1: Lucy (1999) with SEDust

Parameter	Default	Description
`use_dustemis`	.false.	Compute dust emission (requires `par%use_sed`)
`dust_emission_method`	'lucy'	'lucy' (SEDust) or 'bw01' (Bjorkman & Wood)
`dust_model_sed`	'astrodust'	SEDust grain model: 'astrodust', 'dl07', or 'zubko'
`sed_qtable, sed_sizedist`	`SEDust/`	optics / size-distribution paths ('astrodust', 'dl07')
`sed_dl07_sdindex,` `sed_dl07_uisrf`	7, 1.0	'dl07' only: WD01 size-dist. index and reference U
`sed_zubko_config,` `sed_zubko_dir`	`../data/zubko/`	'zubko' only: ZDA config + DustEM data dir

Parameter	Default	Description
`sed_workdir`	blank (auto)	SEDUST data dir; blank auto-resolves to `<MoCafe.x dir>/SEDust/sed`
`sed_NT, sed_Tlo, sed_Thi`	200, 2.7, 5000	Grain temperature grid
`dust_niter`	1	Lucy iterations (dust self-absorption)
`dust_no_photons`	1e6	Dust-emission photons per iteration
`dust_tol`	1e-3	Convergence on the total emitted luminosity

The `'lucy'` method requires `par%save_jlam`. For each cell the local mean intensity J_λ (converted to SI and resampled onto the SEDUST wavelength grid) is passed to SEDUST, which returns the emission spectrum from the full grain-size distribution — equilibrium grains plus stochastically heated small grains and PAH features. The *shape* is taken from SEDUST while the bolometric luminosity is set to the locally absorbed power,

$$L_{\rm emit}({\rm cell}) \;=\; L_{\rm abs}({\rm cell}) \;=\; \rho\kappa \sum_{\lambda} s_{\rm ext}(\lambda)\,[1-\varpi(\lambda)]\,\Big(\textstyle\sum_{\rm paths} w\,L_{\rm pk}\,d\ell\Big)_{\lambda,{\rm cell}}, \tag{5}$$

so total energy is conserved by construction (radiative equilibrium). With `par%dust_niter` > 1 the re-emitted photons are transported so that dust self-absorption heats other cells; the iteration converges when the total emitted luminosity stops changing (needed at high infrared optical depth).

Emission-solver speed options. The exact stochastic solve costs ~ 0.1 s per cell; two faster paths are available:

Parameter	Default	Description
`dust_single_teq`	.false.	One mixture equilibrium T per cell; fastest ($\sim$55$\times$); true $T_{\rm eq}$; *no PAH/stochastic features*
`dust_fast_table`	.false.	Interpolate emission vs. field intensity U over a fixed reference shape; $\sim$1% SED-shape error; keeps stochastic/PAH
`dust_nU`	40	U-grid size for the table path

5.4 Mode 2: Bjorkman & Wood (2001)

Set `par%dust_emission_method` = `'bw01'`. This is an approximate, equilibrium, mixture mean-opacity method with *no iteration*: fixed-energy packets random-walk through scatter and absorb+immediate-reemit events until they escape. Each absorption raises the local temperature (from the running absorbed power) and the packet wavelength is resampled from the temperature-correction spectrum $\propto \kappa_{\rm abs}\,\partial B_\lambda/\partial T$, so the time-averaged emission equals $\kappa_{\rm abs} B_\lambda(T)$ without iterating. Mode 2 needs only `par%kext_file` (not SEDUST) and does not reproduce PAH or stochastic-heating features. It writes `<base>_bwdust` (equilibrium $T_{\rm dust}$ and absorbed-power maps); the emergent dust emission is added to the observer `Scattered` SED image.

5.5 Multiple stellar populations and galaxy models

Parameter	Default	Description
`nsource`	1	Number of stellar components (> 1 activates multi-source)
`src_geometry(i)`	'point'	Geometry of each source (list below)
`src_tstar(i)`	-999	Planck temperature of each source [K]
`src_spectrum(i)`	' '	Spectrum file of each source (overrides src_tstar)
`src_lum(i)`	-999	Luminosity of each source [erg/s] (equal split if unset; derived from the file integral for an absolute spectrum_type file)
`src_x/y/z(i)`	0	Position of each source ('point')
`src_zscale(i)`	-999	Vertical scale height (exponential/sech/exp_spiral)
`src_rscale(i)`	-999	Radial scale length of the exponential disk
`src_sersic_index(i)`	4	Sersic index n of a 'sersic' bulge
`src_reff(i)`	-999	Effective radius (sersic) / scale (boxy/bar/xbar)
`src_axial_ratio(i)`	1	Vertical flattening of a sersic/boxy/bar/xbar bulge
`src_boxiness(i)`	2	Boxiness of a boxy/bar/xbar bulge (2 = ellipsoid, >2 = boxy)

Source geometries. `point`, `uniform`, `gaussian` as before; `exponential` (radially exponential disk $r \sim r e^{-r/r_s}$ $\times$ vertical exponential); `sech` (radial disk $\times$ sech^2); `exp_spiral` (exponential disk with log-spiral arms); `sersic` (3-D deprojected Sersic bulge, radius *alias-sampled* from the analytic cumulative); `boxy/bar/xbar` (generalized-ellipsoid bulges, boxiness `src_boxiness`).

Spiral arms and vertical dust profile (shared by the `exp_spiral` sources and the cylinder dust density):

Parameter	Default	Description
`spiral_m`	0	Number of log-spiral arms (0 = axisymmetric)
`spiral_pitch`	15	Pitch angle [deg]
`spiral_amp`	0.5	Arm amplitude in $1+\mathrm{amp}\sin(m(\ln r/\tan p-\phi))$
`density_zprofile`	'exp'	Vertical dust: 'exp' $e^{-\lvert z\rvert/z_s}$ or 'sech' $\mathrm{sech}^2(z/z_s)$

Sources are sampled in proportion to their luminosity, and `par%luminosity` becomes $\sum_i$`src_lum(i)`. Multiple internal sources work in both the monochromatic and the SED modes. Examples: a disk + bulge SED (`examples/galaxy/galaxy_sed.in`), a face-on spiral galaxy (`spiral_faceon.in`), and the Milky-Way all-sky (`examples/milkyway/mw_allsky.in`).

5.6 HEALPix all-sky interior observer (Milky Way)

Parameter	Default	Description
allsky	.false.	Observer inside the medium; bin every event by sky direction
allsky_x/y/z	0	Interior observer position (code units)
allsky_nside	64	HEALPix resolution (power of 2); $n_{\rm pix} = 12\,n_{\rm side}^2$

Each scatter, dust-reemission, and direct event peels toward the interior observer and is binned into the **HEALPix** pixel (RING scheme, equal-area pixels) of the sky direction it arrives from, following LaRT's `observer_heal` implementation. The event-to-observer optical depth is integrated only over the segment to the observer, using the Cartesian or the octree segment ray trace as appropriate — so the all-sky map works on **both the Cartesian grid and the AMR octree**. The output `<base>_allsky` is a HEALPix ($n_{\rm pix}, \lambda$) surface-brightness map — the diffuse Galactic light and far-infrared cirrus as seen from inside the disk. It carries the standard HEALPix keywords (`PIXTYPE`, `ORDERING='RING'`, `NSIDE`, `NPIX`, `OMEGAPIX`) and a `PixelDir` table of pixel-center unit vectors, so it reads directly with `healpy` (`examples/milkyway/`).

5.7 Modified Random Walk (high optical depth)

Parameter	Default	Description
use_mrw	.false.	Diffusion step in scattering-thick cells
mrw_gamma	2.0	Trigger threshold on the nearest-wall scattering optical depth

In a cell whose nearest-wall *scattering* optical depth exceeds `par%mrw_gamma`, MoCafe replaces the many small steps with a single diffusion step to the inscribed-sphere surface (Min et al. 2009; Robitaille 2010): it samples the total path length inside the sphere from the diffusion first-passage distribution, deposits the absorption-weighted path into the J_λ tally, and decays the weight by $\exp(-\alpha_{\rm abs}\,ct)$. The Lucy transport also applies Russian roulette so high-albedo, high-τ clouds do not diffuse indefinitely. MRW is exact in absorbing dust (where it triggers rarely); its speed-up is large only when individual cells are very thick ($\tau_{\rm cell} \gg 1$) and nearly pure-scattering.

5.8 Dust emission on the AMR octree

The entire dust-emission suite — SED transport, the J_λ tally, Mode 1 (Lucy, including the single-$T_{\rm eq}$ and table options), Mode 2 (Bjorkman & Wood), and the Modified Random Walk — runs on the AMR octree (`par%grid_type = 'amr'`) exactly as on the Cartesian grid. A "cell" is simply an octree leaf: the radiation field, absorbed power, and dust temperature are tallied per leaf (code-volume and opacity of each leaf), and MRW uses the nearest leaf-face distance for its trigger. Set up the octree as usual (`par%grid_type = 'amr'`, `par%amr_file`, and `par%dust_model` for the leaf opacity) and add the emission parameters above. The derived files carry arrays indexed by leaf instead of Cartesian cubes: `<base>_jlam`, `<base>_dustsed`, and `<base>_bwdust` each add a `LeafXYZ` table of leaf-center coordinates so the leaf quantities can be placed in space. Example: `examples/amr_sphere/amr_dustemis.in` (a uniform level-5 octree sphere) reproduces the energy budget and SED of the equivalent Cartesian sphere.

5.9 A minimal emission run

```
&parameters
 par%no_photons  = 2.0e6
 par%use_sed     = .true.
 par%save_jlam   = .true.
 par%use_dustemis    = .true.
 par%dust_single_teq = .true.     ! fast equilibrium solve
 par%luminosity  = 3.828e33       ! erg/s
 par%nlambda     = 100
 par%lambda_min  = 0.0912
 par%lambda_max  = 2000.0
 par%kext_file   = 'data/kext_astrodust_MW.dat'
 par%tstar       = 1.0e4
 par%taumax      = 5.0
 par%source_geometry = 'point'
 par%distance_unit   = 'pc'
 par%rmax = 1.0
 par%nx = 33
 par%ny = 33
 par%nz = 33
 par%distance = 1.0e5
/
```

6. Output Products

`par%file_format` is authoritative: if `par%out_file`'s extension disagrees, MoCafe rewrites it (with a warning) so that the main file and all derived files share one container. HDF5 mirrors the FITS HDU layout (each image HDU becomes a group named after its `EXTNAME` with a `data` dataset; FITS keywords become group attributes).

Suffix	Written when	Contents
`_obs`	always	Observer images; in SED mode `Scattered`/`Direct` become (x, y, λ) cubes plus the wavelength / optics tables
`_jlam`	save_jlam	$J_\lambda(\lambda, x, y, z)$, $J_{\rm bol}$, energy-conservation keywords
`_dustsed`	use_dustemis ('lucy')	Emergent and intrinsic dust SED, a pixel-resolved `DustEmis_image`(x, y, λ), and $T_{\rm dust}/L_{\rm dust}$ maps for each cell
`_bwdust`	'bw01'	Equilibrium $T_{\rm dust}$ and absorbed-power maps
`_allsky`	allsky	HEALPix $(n_{\rm pix}, \lambda)$ all-sky surface-brightness map (RING) + `PixelDir`
`_obs_tau`	sightline_tau	Sight-line optical-depth map

Suffix	Written when	Contents
`_stokes`	use_stokes	Stokes I, Q, U, V images

On the AMR octree the `_jlam`, `_dustsed`, and `_bwdust` images are arrays indexed by leaf rather than Cartesian cubes, and each adds a `LeafXYZ` table of leaf-center coordinates. Read and convert the outputs with `python/mocafe_io.py` (`info`, `peek`, `convert`).

7. Worked Examples

Directory	Shows
`examples/point_source, uniform_source, external_*`	Cartesian grids and source geometries
`examples/mono_multi_source`	Monochromatic scattering with two internal point sources (`par%nsource` > 1)
`examples/compose_int_ext`	Internal source + external field composed in one run (`compose_mono.in`, `compose_sed.in`)
`examples/agtau_list`	(a, g) and τ single-run scans
`examples/clump_sphere, amr_sphere, amr_tng, amr_ramses`	Clumpy and AMR media
`examples/dustemis`	Dust emission: Lucy+SEDUST, iteration, single-$T_{\rm eq}$, table, B&W, MRW
`examples/sed_external`	External-field SED illumination (Planck shape + `par%ext_intensity`; absolute `per_um` J file)
`examples/sed_multi_source`	Multi-source SED with luminosity derived from absolute spectra
`examples/amr_sphere`	Dust emission on the AMR octree (`amr_dustemis.in`; matches the Cartesian sphere)
`examples/multipop`	Two stellar populations (hot young + cool old)
`examples/galaxy`	Galaxy SED (`galaxy_sed.in`: disk + Sersic bulge) and a face-on spiral galaxy (`spiral_faceon.in`)
`examples/milkyway`	All-sky map from an interior observer (spiral dust disk)
`examples/benchmarks`	Camps (2015) SHG dust-emission benchmark and Mode 1/2 τ sweep

The `python/` tools read and plot these outputs: `mocafe_io.py` (reader), `plot_dust_sed.py`, `plot_dust_maps.py`, `plot_jlam.py`, `plot_allsky.py`, `plot_galaxy_sed.py`, and the `quickstart_dustemis.ipynb` notebook.

Figure 1: *Left*: the UV-to-FIR SED of the `examples/galaxy` model (exponential disk + Sersic bulge): the attenuated starlight and the thermal dust emission (PAH bands and a far-infrared peak), with $L_{\rm dust}/L_\star \approx 0.8$. *Right*: the mid-plane dust-emission luminosity of the `spiral_faceon.in` model, showing the $m = 2$ log-spiral arms imposed on the dust density.

8. Validation

- **Regression.** With the emission features off, the monochromatic scattering, the $(a, g)/\tau$ scans, and the clumpy-medium outputs are bit-identical to a stand-alone scattering run.
- **Dust-emission engine (cross-code).** Run through the SEDUST library with the Zubko et al. (2004) BARE-GR-S model, the emission spectra agree with the Camps et al. (2015) seven-code stochastic-heating benchmark—which includes SKIRT and DIRTY—to a median relative difference of 2–4% across $U = 10^{-2} \ldots 10^4$, and the emission peak matches the code median to within one wavelength bin.
- **Energy conservation.** The intrinsic dust SED integrates to the absorbed luminosity to machine precision (the emission is normalized to the locally absorbed power). The Lucy path-length estimator and the Bjorkman & Wood immediate-reemission scheme agree on the absorbed luminosity to $< 2\%$ over the sweep $\tau_V = 0.1 \ldots 20$.
- **AMR vs. Cartesian.** Dust emission on a uniform level-5 octree sphere reproduces the reprocessed fraction (0.863 at $\tau_V = 5$) and energy balance of the equivalent Cartesian sphere; Bjorkman & Wood and MRW on the octree conserve energy identically.

These checks are consolidated in `examples/benchmarks/validate.py`, which re-runs the energy-conservation, two-method, and SHG cross-code tests and reports PASS/FAIL. External full-galaxy transport benchmarks (the TRUST slab, a direct SKIRT/DIRTY image cross-check) need those codes or their published reference solutions and are added as they become available; the emission engine is already cross-checked against SKIRT and DIRTY through the SHG benchmark above.

9. Algorithm Notes

Packet energy. Each packet carries a physical luminosity $L_{\rm pk}$ [erg/s]: $L_\star/N_\star$ for stellar packets and $L_{\rm dust}/N_{\rm dust}$ for dust-emission packets, so populations with different packet energies combine correctly in the J_λ tally.

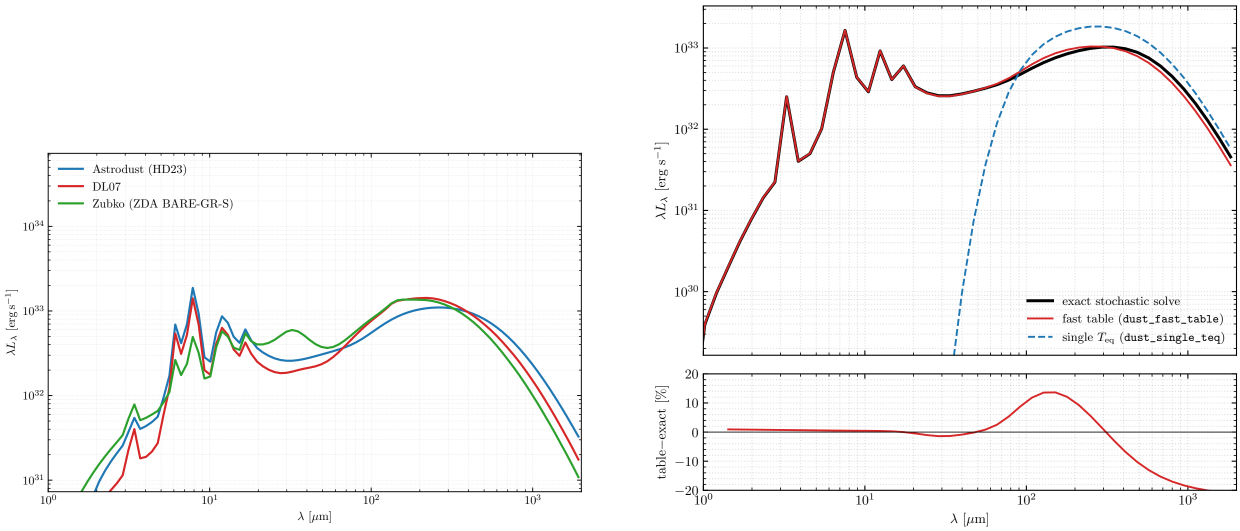

Figure 2: *Left*: the emission spectrum of a single cell heated by the same radiation field, for the three SEDUST grain models (`astrodust`, `dl07`, `zubko`); the total power is identical (each is normalized to the absorbed luminosity) while the PAH bands and the far-infrared slope differ by model. *Right*: the `par%dust_fast_table` approximation against the exact stochastic solve — the total luminosity is identical, the PAH/mid-infrared region agrees to $\sim 1\%$, and the largest deviations sit on the far-infrared slope.

Lucy iteration. The stellar J_λ is tallied once and reused; each iteration transports fresh dust-emission photons (sampled per cell from the emission spectrum), adds their J_λ to the stellar field, and recomputes the emission. The total emitted luminosity obeys $L^{(k)} = L^{\star}_{\rm abs} + f\,L^{(k-1)}$ with f the reabsorbed fraction, a geometric series converging to $L^{\star}_{\rm abs}/(1-f)$.

Bjorkman & Wood temperature. A cell's equilibrium temperature is found by inverting the monotonic $\kappa_B(T) = \int \kappa_{\rm abs}(\lambda) B_\lambda(T)\,d\lambda$ against the running absorbed power per H nucleus.

References

- Lucy, L. B. 1999, A&A, 344, 282 — iterative mean-intensity / dust-emission estimator.
- Bjorkman, J. E. & Wood, K. 2001, ApJ, 554, 615 — immediate reemission with temperature correction.
- Camps, P. et al. 2015, A&A, 580, A87 — stochastic-heating dust-emission benchmark.
- Zubko, V., Dwek, E., & Arendt, R. G. 2004, ApJS, 152, 211 — BARE-GR-S dust model.
- Min, M. et al. 2009, A&A, 497, 155; Robitaille, T. P. 2010, A&A, 520, A70 — Modified Random Walk.
- Seon, K.-I. 2010, PKAS, 25, 177 — the (a,g) single-run scan.
- Jonsson, P. 2006, MNRAS, 372, 2 — the polychromatic optical-depth scan.
- Górski, K. M. et al. 2005, ApJ, 622, 759 — HEALPix pixelization (all-sky interior observer).
- Amanatides, J. & Woo, A. 1987, Eurographics '87 — fast voxel traversal.

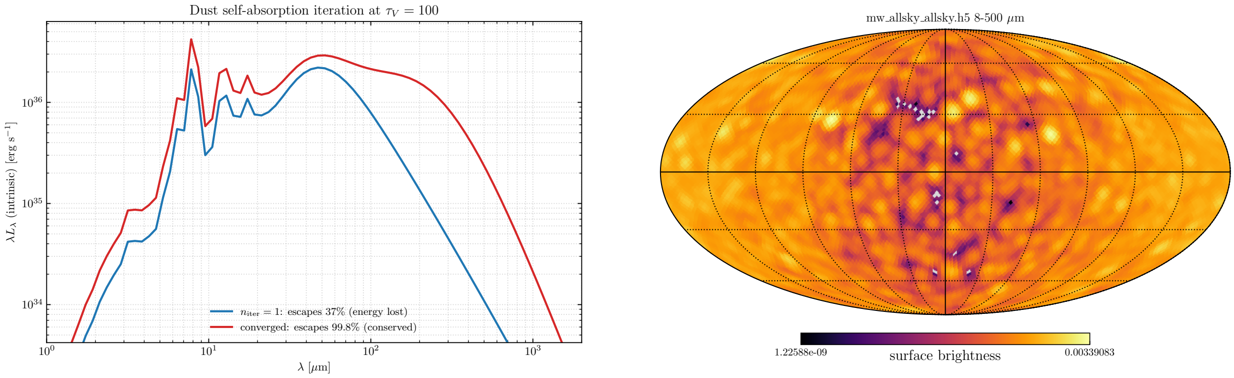

Figure 3: *Left*: convergence of the dust self-absorption iteration (`par%dust_niter`) at $\tau_V = 100$: the emitted luminosity converges geometrically, and only the converged run conserves energy (a single pass loses the reabsorbed dust emission). *Right*: the smoothed 8–500 μm all-sky map from the interior-observer Milky-Way model (`examples/milkyway`), a HEALPix Mollweide projection showing the Galactic-plane concentration.

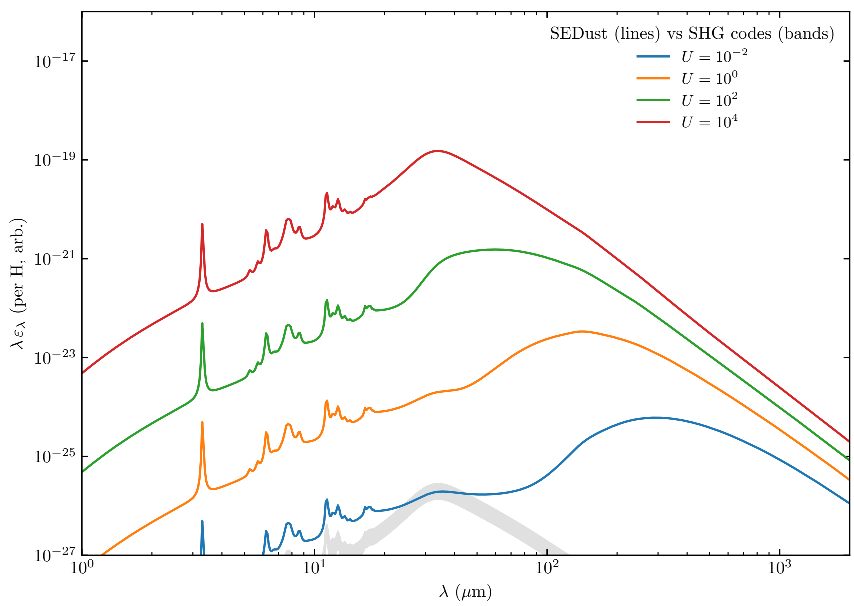

Figure 4: The SEDust single-cell emission spectrum (zubko BARE-GR-S) against the Camps et al. (2015) seven-code stochastic-heating benchmark at several field strengths U; the shaded band is the code-to-code spread. MoCafe tracks the benchmark median to a 2–4% relative difference over $U = 10^{-2} \ldots 10^{4}$.